Comparative Statistical Power of N-of-1 Trials versus Randomized Controlled Trials: A Simulation Study of Prazosin for PTSD Nightmares

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August 15, 2026

Abstract

Background: N-of-1 trials offer a promising approach for precision medicine by evaluating treatment effects within individual patients through repeated crossover designs. Their statistical properties relative to traditional randomized controlled trials remain incompletely characterized across clinically realistic effect-size and carryover regimes. **Methods:** We conducted an ADEMP-compliant Monte Carlo simulation study comparing the statistical power of aggregated N-of-1 hybrid trials versus parallel-group RCTs for detecting treatment effects of prazosin in reducing PTSD nightmares. The N-of-1 design enrolled 35 individuals each completing 8-period crossover trials analyzed by a linear mixed-effects model with continuous-time AR(1) residuals; the RCT arm used ANCOVA at matched sample size $N = 35$. Carryover-contaminated placebo periods were excluded from the N-of-1 analysis before fitting. We ran $n_{\text{sim}} = 2{,}000$ replicates per cell, uniform across all primary and sensitivity analyses. **Results:** The aggregated N-of-1 hybrid dominated the parallel-group RCT across the full effect-size range. At a true effect of -2.0 nightmares per week, N-of-1 power was 1.000 versus 0.675 (Monte Carlo standard error, MCSE, 0.010) for the RCT; at -1.0, N-of-1 power was 0.830 (MCSE 0.008) versus 0.230 (MCSE 0.009). Bias was negligible in both designs. Coverage of nominal-95% CIs was 0.940–0.949 for the N-of-1 arm and 0.939 for the RCT arm across all effect sizes. **Conclusions:** Aggregated N-of-1 hybrid trials achieve substantially higher statistical power than parallel-group RCTs at matched sample size when within-subject carryover is managed by period exclusion. The efficiency advantage is largest in the small-to-moderate effect-size regime where parallel-group RCTs are essentially underpowered. **Keywords:** N-of-1 trials; randomized controlled trials; statistical power; crossover design; carryover effects; PTSD; prazosin; precision medicine.

Contents

1 Abstract

Background. Aggregated N-of-1 trial designs and parallel-group randomized controlled trials are the two principal methodological alternatives for evaluating treatment effects under substantial individual heterogeneity. The published literature characterizes the variance-component foundations of each design separately, but has not, to the best of our knowledge, delivered a comprehensive sample-size and power comparison across clinically realistic effect-size and carryover regimes that admits Monte Carlo standard-error reporting within the ADEMP framework of Morris, White, and Crowther [-@morris2019simulation].

Methods. We conduct an ADEMP-compliant Monte Carlo simulation study comparing the aggregated N-of-1 hybrid design to a parallel-group RCT at matched sample size $N = 35$, across a grid of true-effect sizes $\{0, -0.5, -1.0, -2.0\}$ nightmares per week, calibrated to the prazosin/PTSD reference application. The hybrid N-of-1 arm fits an ‘nlme::lme‘ model with continuous-time AR(1) residual correlation; the parallel-group RCT arm fits a baseline-adjusted linear model (ANCOVA). Carryover-contaminated placebo periods are excluded before fitting the N-of-1 model. Each cell is run at $n_{\text{sim}} = 2{,}000$ replicates; null cells (true effect zero) provide explicit Type I error checks. Estimands include power, empirical effect-size bias, empirical SE, model SE, and nominal-95are reported with Monte Carlo standard errors. Sensitivity sweep S1 examines carryover half-lives in $\{0.5, 1, 3, 5, 7\}$ days.

Results. The aggregated N-of-1 hybrid dominated the parallel-group RCT across the entire effect-size range. At true effect -2.0, N-of-1 power was 1.000 versus parallel-group 0.675 ± 0.010; at -1.0, 0.830 ± 0.008 versus 0.230 ± 0.009; at -0.5, 0.311 ± 0.010 versus 0.104 ± 0.007. The two power curves did not cross within the simulated range; gaps exceeded five combined MCSEs at every non-null effect size. The relative N-of-1-to-RCT power ratio was approximately $3.0\times$ at the moderate-effect -0.5 cell. Bias was negligible across all cells for both designs. Coverage of nominal-95the RCT arm, each within two MCSEs of the nominal level. The N-of-1 analysis was insensitive to the true carryover half-life, confirming robustness of the period-exclusion strategy. Type I error was 0.051 ± 0.005 for the N-of-1 hybrid and 0.061 ± 0.005 for the RCT; the RCT rate is marginally above nominal, consistent with small-sample effects in lm at $N = 35$.

Conclusions. Aggregated N-of-1 hybrid trials achieve substantially higher statistical power than parallel-group RCTs at matched sample size when individual variability is substantial and within-subject carryover is manageable. The efficiency advantage is largest in the small-to-moderate effect-size regime

where parallel-group RCTs are essentially powerless. Trial designers in precision-medicine settings with high between-patient variability should consider the aggregated N-of-1 design as the preferred default.

2 Introduction

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Clinical research has moved steadily toward precision-medicine approaches that recognize substantial between-patient variance in treatment response[1,2]. Traditional randomized controlled trials (RCTs), while remaining the standard for establishing population-level treatment effects, may not capture the between-patient variability that is central to precision-medicine therapeutic decision-making[3,4].

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N-of-1 trials, also known as single-patient crossover trials, evaluate treatment efficacy within individual patients through repeated crossover periods[5,6]. In these designs each patient serves as their own control, which reduces the confounding effect of between-patient variability while providing individual-level treatment-effect estimates[7,8]. When appropriately aggregated across patients, N-of-1 trials can support population-level inference while simultaneously informing individual treatment decisions[9,10].

2.1 Methodological Considerations in N-of-1 Trial Design

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The statistical analysis of N-of-1 trials presents methodological challenges that differ substantially from those of traditional parallel-group designs[11,12]. Among the key considerations are the appropriate handling of carryover effects, period effects, and temporal trends within individual patients[13,14]. Carryover effects, representing the residual influence of treatment from previous periods, are particularly critical in crossover designs and can substantially affect both the validity and the power of statistical analyses[15,16].

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The aggregation of individual N-of-1 trials to yield population-level estimates requires meta-analytic approaches that account for both within-patient and between-patient variability[7]. Random-effects meta-analysis methods, such as the DerSimonian-Laird approach, can appropriately combine individual treatment-effect estimates while incorporating heterogeneity between patients[17].

2.2 Sample Size and Power Considerations

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Sample size determination for N-of-1 trials involves the number of patients, the number of crossover periods per patient, and the expected magnitude of within-patient relative to between-patient variability[18]. Simulation studies have suggested that N-of-1 designs may achieve adequate statistical power with smaller total sample sizes than parallel-group RCTs, particularly when treatment effects are moderately large and individual variability is substantial[19]. But how large is the advantage, and how does it depend on the carryover regime? The relative statistical efficiency of N-of-1 trials compared to RCTs remains context-dependent and has not been comprehensively evaluated across varying effect sizes and carryover scenarios[9], and understanding these relationships is important for informed study design selection.

2.2.1 Analytic Frameworks for Power in N-of-1 Designs

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The statistical power of an N-of-1 trial depends on a fundamentally different set of variance components than a parallel-group RCT. Because each patient serves as their own control, the between-patient variance $\sigma^2_{\text{between}}$ is eliminated from the treatment-effect estimator, and the relevant sources of uncertainty become the within-patient, between-period variance σ^2 and the residual measurement error[11,20].

2.2.1.1 Variance decomposition and the paired-cycles model

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Senn[20] formalized the sample size problem for sets of N-of-1 trials under a Normal-Normal mixture model. Suppose n patients are each randomized in k cycles of paired treatment periods, each cycle containing one period of treatment A and one of treatment B in random order, and let d_{ij} denote the within-cycle treatment difference for patient i in cycle j. The model assumes

$$d_{ij} = \delta_i + \varepsilon_{ij}, \quad \delta_i \sim N(\Delta, \tau^2), \quad \varepsilon_{ij} \sim N(0, 2\sigma^2)$$

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where Δ is the population mean treatment effect, τ^2 is the between-patient variance of individual treatment effects (the treatment-by-patient interaction), and $2\sigma^2$ is the within-patient, within-cycle variance of the paired differences[20]. The factor of 2 arises because d_{ij} is the difference of two independent within-period observations, each with variance σ^2.

Under this model the patient-level mean difference $\bar{d}_{i\cdot} = k^{-1}\sum_j d_{ij}$ has variance $\tau^2 + 2\sigma^2/k$, and the grand mean $\hat{\Delta} = n^{-1}\sum_i \bar{d}_{i\cdot}$ has variance

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + 2\sigma^2/k}{n}$$

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This expression is central to the trade-off between the number of patients n and the number of cycles per patient k: increasing k reduces only the $2\sigma^2/k$ component, whereas increasing n reduces the entire expression proportionally[7,20].

2.2.1.2 Closed-form power for fixed- and random-effects analyses

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Senn[20] distinguished two inferential goals with distinct power formulas. For a *random-effects* analysis targeting the population treatment effect Δ, the patient-level summary-measures approach yields a one-sample t-test on the n patient means $\bar{d}_{i\cdot}$ with $n-1$ degrees of freedom; the variance at the patient level is $\tau^2 + 2\sigma^2/k$, estimated directly from the n observed summary measures. Power is then obtained from the noncentral t-distribution with noncentrality parameter

$$\lambda_{\mathrm{RE}} = \frac{\Delta}{\sqrt{(\tau^2 + 2\sigma^2/k)/n}}$$

and $n-1$ degrees of freedom.

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For a *fixed-effects* analysis, appropriate when the inferential goal is to test whether the treatments differ for the patients actually studied, the treatment-by-patient interaction τ^2 is excluded from the variance estimator. That is, the relevant variance of $\hat{\Delta}$ is $2\sigma^2/(nk)$, estimated with $n(k-1)$ degrees of freedom obtained by pooling within-patient sums of squares. Power follows from the noncentral t-distribution with

$$\lambda_{\mathrm{FE}} = \frac{\Delta}{\sqrt{2\sigma^2/(nk)}}$$

and $n(k-1)$ degrees of freedom[20]. Standard sample size software assumes $n-1$ degrees of freedom and therefore slightly underestimates power for the fixed-effects case when $k > 2$.

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Senn and Julious[12] provided a complementary tutorial demonstrating how these paired-cycle differences are computed and analyzed in practice, including the estimation of the pooled within-patient standard deviation $\hat{\sigma}_w = \sqrt{\sum_i \mathrm{SS}_i / \sum_i (m_i - 1)}$ and the construction of shrinkage (BLUP) estimators for individual patient effects.

2.2.1.3 The n-versus-k trade-off in series of N-of-1 trials

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Zucker et al.[7] studied the trade-off between the number of patients n and the number of paired treatment periods k using a random-effects meta-analysis framework. Their precision formula,

$$\mathrm{Var}(\hat{\Delta}) = \frac{\tau^2 + \sigma_w^2/k}{n}$$

where σ_w^2 denotes the within-patient variance of paired differences, shows that the marginal benefit of additional cycles diminishes rapidly once $\sigma_w^2/k \ll \tau^2$. When between-patient heterogeneity dominates (that is, $\tau^2 \gg \sigma_w^2$), additional patients yield greater precision gains than additional cycles; conversely, when within-patient noise is large relative to heterogeneity, repeated cycles on the same patient remain valuable.

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Arends et al.[18] extended these results to derive explicit sample-size formulas for series of N-of-1 trials analyzed via random-effects meta-analysis, providing tables for jointly determining n and k as a function of Δ, σ_w^2, and τ^2.

2.2.1.4 Complications motivating simulation

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The closed-form solutions above rest on several simplifying assumptions that may not hold in practice. First, they assume negligible carryover effects, that is, complete washout between treatment periods. When carryover is partial, as may occur with prazosin's neuroadaptive effects on sleep architecture, the effective treatment contrast is attenuated and variance components shift in ways not captured by balanced-design formulas[14].

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Second, within-patient observations are typically serially correlated, and ignoring this correlation can inflate Type I error or distort power estimates[13]. Wang and Schork[21] demonstrated via likelihood-ratio methods under an AR(1) correlation structure that serial correlation can substantially reduce power, particularly in designs with few, long treatment periods, and they showed that more frequent period alternation mitigates this power loss at the cost of practical feasibility.

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Third, practical N-of-1 trials may feature unbalanced designs with unequal period lengths, missing periods, or adaptive stopping rules that violate the balanced paired-cycles assumption. Fourth, the outcome of interest (e.g., nightmare frequency) may be better modeled as count data, whereas the closed-form formulas assume normality. Fifth, when the inferential goal encompasses both individual-level treatment decisions and population-level inference, the power trade-off between n and k has no simple closed-form solution because both the shrinkage estimator and the population estimate must be jointly optimized[20].

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These complications collectively motivate the Monte Carlo simulation approach adopted in the present study, which permits evaluation of power under realistic combinations of carryover magnitude, variance structure, and design imbalance.

2.3 Single-Case Experimental Design Framework

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N-of-1 trials represent a specific application of single-case experimental designs (SCEDs) that have been extensively developed in behavioral and educational research[22,23]. The methodological rigor of SCEDs has evolved to include analytical approaches that can provide valid causal inferences from individual-level data[24,25].

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Recent developments in SCED methodology have emphasized the importance of proper randomization, adequate baseline measurement, and appropriate statistical analysis[26]. These advances have strengthened the evidence base for using N-of-1 trials as a complement to traditional group-based designs[27].

2.4 Clinical Application: Prazosin for PTSD

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Post-traumatic stress disorder (PTSD) represents a particularly instructive clinical context for evaluating N-of-1 trial methodology, given the substantial individual variation in treatment response and the chronic, fluctuating nature of symptoms[28]. Prazosin, an α_1-adrenergic receptor antagonist, has demonstrated efficacy for reducing trauma-related nightmares in multiple RCTs[29,30], though individual response patterns vary considerably.

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The pharmacokinetic properties of prazosin, including its relatively short half-life, make it particularly suitable for crossover designs while also necessitating careful consideration of carryover effects[31,32]. Previous studies have documented both the efficacy and the individual variability of prazosin response[33,34], providing an empirical foundation for the simulation parameters we employ here.

2.5 Regulatory and Implementation Considerations

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The increasing recognition of N-of-1 trials by regulatory agencies[35] and research institutions[2] has highlighted the need for robust methodological guidelines and statistical frameworks. Professional organizations have developed comprehensive recommendations for N-of-1 trial design and implementation[6,36], while specialized software tools have emerged to support N-of-1 trial analysis[25,37].

2.6 Prior Simulation Comparisons of N-of-1 and Parallel Designs

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The comparative efficiency of N-of-1 and parallel-group designs has been examined in several simulation studies that form the immediate methodological precedent for the present work. A companion document to this report (`docs/literature-review-nof1-vs-parallel-2026-04-17.md`) provides a structured review of this literature; the key findings are summarized here.

2.6.1 Direct head-to-head simulation comparisons

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The most frequently cited direct comparison is Blackston and colleagues' study of aggregated N-of-1 trials versus parallel and crossover RCTs[19]. Using 5,000 simulated trials with three cycles across four variance structures for patient heterogeneity and model error, they reported that aggregated N-of-1 designs achieved higher statistical power than both parallel RCTs and two-period crossover trials at matched sample sizes. They also found that Type I error could be inflated under N-of-1 when moderate-to-strong carryover or washout was present but not modeled, and that patient-level random effects were recoverable in N-of-1 but not in the parallel RCT, because the latter provides only one observation per participant.

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A more recent comparison by Carrozzo and colleagues[38] examined a two-arm parallel RCT, a two-period crossover, and a meta-analysis of N-of-1 trials for a cardiovascular digital-health intervention. Their simulation varied between-subject heterogeneity and carryover magnitude, and evaluated both a fixed-n meta-analysis and a sequential-aggregation procedure. They reported that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm designs across most heterogeneity and carryover settings, and that the sequential variant crossed the 80% threshold earlier than the fixed-n procedure.

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The specific methodological ancestor of the biomarker-interaction framework used in the present sensitivity analyses is Hendrickson and colleagues' simulation of aggregated N-of-1 designs for predictive-biomarker validation[39]. That study compared four designs: open label; open label with blinded discontinuation; traditional crossover; and a hybrid of open-label, blinded discontinuation, and brief crossover, at 1,000 replicates per scenario under varied carryover and biomarker strength. The hybrid design provided power close to the traditional crossover while preserving an initial open-label titration phase and showing smaller power erosion under carryover.

2.6.2 Analytical complications and misspecification

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Wang and Schork[21] extended the analytical and simulation treatment of power to cases with AR(1) serial correlation, heteroscedastic responses, multiple evaluation periods, and washout periods. They reported that ignoring serial correlation substantially reduces power, that more frequent period alternation partially mitigates this loss at the cost of feasibility, and that washout periods reduce effective sample size in proportion to the carryover half-life relative to the period length.

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Gaertner and colleagues[40] simulated aggregated N-of-1 trials under directed acyclic graphs that encode carryover and treatment-dependent covariate structures. They compared linear mixed models, Bayesian networks, G-estimation, and a carryover-adjusted parametric model, and reported that all methods perform well without carryover; under carryover, the carryover-adjusted parametric model is unbiased while the other methods show bias. Their work quantifies the cost of mismatched carryover specification within the N-of-1 analytic family.

2.6.3 Closed-form and semi-analytical frameworks

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Senn formalized sample-size planning for sets of N-of-1 trials under a Normal-Normal mixture model and identified five planning criteria with corresponding closed-form solutions[20]. The decomposition $\mathrm{Var}(\hat{\Delta}) = (\tau^2 + 2\sigma^2/k)/n$ makes explicit the trade-off between the number of patients n and the number of cycles per patient k: increasing k reduces only the $2\sigma^2/k$ component, whereas increasing n reduces the entire expression proportionally. Zucker, Ruthazer, and Schmid[7] compared summary and individual-participant-data meta-analyses, repeated-measure models, Bayesian hierarchical models, and period or pair crossover models on a published N-of-1 series, and characterized the within- and between-patient variance structure that determines relative efficiency.

2.6.4 Recent design refinements

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Recent work extends the comparison in three directions. Jiang and colleagues[41] proposed sequential monitoring rules for both single-person and multi-person N-of-1 trials, with sample-size calculations in an Alzheimer's disease motivating application. Selukar, Prince, and May[42] proposed sequential monitoring at the per-trial level with inverse-variance random-effects combination across trials, and reported that Type I error can be substantially inflated under a small number of planned blocks but reaches nominal rates with more blocks or alpha-spending corrections. Giai and colleagues[43] evaluated generalized pairwise comparison (Net Benefit) estimators in N-of-1 series, and reported that individual-level Net Benefit estimation requires very long observation series to be adequately powered.

2.6.5 Summary

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Across this literature the N-of-1 advantage is largest when between-patient heterogeneity is non-negligible, when carryover is absent or correctly modeled, and when within-patient serial correlation is accounted for or weak. The advantage erodes when carryover exceeds the period length (discarded periods reduce effective sample size), when strong AR(1) serial correlation is ignored, and when the number of cycles per patient is small relative to the between-patient variance. Type I error is more sensitive to analytic misspecification under N-of-1 than under parallel designs, and interaction-target analyses (biomarker-treatment) have qualitatively different power profiles than average-effect analyses. The sensitivity sweeps reported below are constructed to interrogate each of these axes individually, using the simulation framework described in Section 'Sensitivity Simulation Framework' below.

2.7 Objectives

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The primary objective of this study was to compare the statistical power of aggregated N-of-1 trials versus traditional parallel-group RCTs for detecting treatment effects, using prazosin for PTSD nightmares as a clinically calibrated reference application. Secondary objectives included: (1) evaluating the impact of carryover effects with varying pharmacokinetic half-lives on statistical power; (2) examining the robustness of the period-exclusion carryover-management strategy; and (3) identifying the conditions under which the N-of-1 approach may be preferred over traditional parallel-group designs.

3 Methods

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We follow the ADEMP framework of Morris, White, and Crowther[44]: Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. In this section we shall describe each component in turn.

3.1 Estimands and pre-registration

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The **primary estimand** is the average treatment-effect coefficient β_D from the linear mixed-effects analysis-model formula `Sx ~ Dbc + t + (1|ptID)` with `corCAR1(form = ~t|ptID)` for the aggregated N-of-1 arm, and the corresponding ANCOVA endpoint coefficient for the parallel-group comparator. The **primary inferential output** is the power $\pi = \Pr(p < 0.05)$ for the two-sided $H_0 : \beta_D = 0$. **Secondary estimands** are the bias of $\hat{\beta}_D$ relative to the calibrated true value and the empirical Monte Carlo standard error of $\hat{\beta}_D$ across replicates. **Performance measures** are reported with binomial-proportion Monte Carlo standard errors $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for proportions and the standard MCSE of the within-replicate mean for continuous quantities. The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`, which fixes the parameter grid for the primary comparison (M1) and the twelve sensitivity sweeps (S1-S12) before the production runs.

3.2 Simulation design and data-generating mechanism

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Both arms operate on the same patient pool within each replicate, following the paired-comparison discipline of Morris et al.[44(sec5.4)]. Each replicate begins by drawing $N = 35$ patient-level baseline nightmare frequencies from Normal(μ_0, σ_τ^2) with $\mu_0 = 8.0$ nightmares per week and $\sigma_\tau = 1.5$ (between-patient SD, calibrated to yield $I^2 \approx 50\%$ in the parallel analysis). This shared pool then drives the N-of-1 arm and the RCT arm independently; the within-replicate covariance between the two power estimates is exploited when reporting paired power differences.

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For the **N-of-1 arm**, each patient completes 8 periods (4 prazosin, 4 placebo) in a balanced random sequence, with each period lasting 7 days. Within-patient residuals are drawn from a zero-mean multivariate normal distribution whose covariance follows the continuous-time AR(1) structure with SD $\sigma_w = 1.0$ and decay rate $\rho = 0.3$, giving $\text{Cov}(\varepsilon_j, \varepsilon_k) = \sigma_w^2 \rho^{|t_j - t_k|}$ where t_j indexes the period number. Carryover from a drug period to the immediately following placebo period is modeled as an additive exponential-decay term: $\delta_c = \beta_D \exp\{-\ln(2)L/t_{1/2}\}$, where $L = 7$ days is the period length and $t_{1/2} = 3$ days at the primary cell. The true effect size grid is $\beta_D \in \{0, -0.5, -1.0, -2.0\}$ nightmares per week.

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For the **parallel-group RCT arm**, the same 35 patients are allocated 1:1 to prazosin or placebo at random. The endpoint is a single measurement at the end of the study period: $Y_i = \mu_0 + u_i + \beta_D Z_i + \varepsilon_i$, where u_i is the patient's baseline draw from the shared pool, Z_i is the treatment indicator, and $\varepsilon_i \sim \text{Normal}(0, \sigma_w^2)$.

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Carryover sensitivity sweep S1 varies $t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days at fixed $\beta_D = -2.0$.

Units. Carryover half-lives $t_{1/2}$ are quoted in **days** throughout this paper, matching prazosin's pharmacokinetic scale. Companion papers in this series that target the biomarker-treatment interaction quote $t_{1/2}$ in **weeks** on the canonical grid $\{0, 0.5, 1.0\}$; the conversion is 1 week = 7 days, so that grid corresponds to $\{0, 3.5, 7\}$ days on the present scale. Timepoints t are weekly in both conventions.

3.3 Statistical analysis models

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N-of-1 arm. Each replicate is analyzed with the linear mixed-effects model

$$Y_{ij} = \beta_0 + \beta_D D_{bc,ij} + \beta_t t_{ij} + u_i + \varepsilon_{ij}$$

where Y_{ij} is the observed symptom score (code column `Sx`), $D_{bc,ij}$ is the continuous exposure-decayed drug indicator (code column `Dbc`), t_{ij} is the within-patient period index, $u_i \sim \text{Normal}(0, \sigma_u^2)$ is the patient random intercept, and ε_i has `corCAR1` covariance. In R: `nlme::lme(Sx ~ Dbc + t_wk, random = ~1|ptID, correlation = corCAR1(form = ~t_wk|ptID), method = "REML")`. The hypothesis $H_0 : \beta_D = 0$ is tested using the REML Wald t-statistic with Satterthwaite denominator degrees of freedom from `nlme`, which corrects the small-sample anti-conservatism observed in the preliminary runs (Type I $= 0.061$ under the pure Wald test at $N = 35$).

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Parallel-group RCT arm. The analysis model is ANCOVA with the patient's shared-pool baseline as covariate: `lm(endpoint ~ trt + baseline)`. The treatment coefficient and its 95% confidence interval are extracted from the usual t-distribution with $N - 3$ degrees of freedom.

3.4 Simulation size and random-number discipline

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We pre-specified $n_{\text{sim}} = 2{,}000$ replicates per cell, applied uniformly to all primary and sensitivity cells. At $n_{\text{sim}} = 2{,}000$ the MCSE on a power estimate near 0.75 is $\sqrt{0.75 \times 0.25/2000} \approx 0.010$ (1.0 percentage points), and on a Type I estimate near 0.05 it is approximately 0.005 (0.5 percentage points); both are below the 1.5-percentage-point target adopted in the pre-registration.

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Random-number discipline follows Morris et al.[44(sec4.6)]. `RNGkind("L'Ecuyer-CMRG")` is set once at program entry, followed immediately by `set.seed(2024L)`. No `set.seed()` calls appear inside simulation functions. Per-replicate `.Random.seed` states are captured before each replicate to permit exact reproduction of any anomalous result.

3.5 Software and reproducibility

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All analyses were conducted in R 4.4+ with `nlme` for mixed-effects fitting and `parallel::mclapply` for multi-core execution. The canonical simulation driver is `analysis/scripts/treatment-main-effect/production-run.R`; output is stored as `analysis/data/derived_data/04-mainfx-production.rds`. Full package dependencies are captured in `renv.lock`.

4 Results

4.1 Headline findings

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At the primary scenario ($N = 35$ per design, true effect $\beta_D = -2.0$ nightmares per week, carryover half-life 3 days, $n_{\text{sim}} = 2000$), the empirical power under the Satterthwaite-corrected Wald test was 67.5% (MCSE 1.0 pp) for the parallel-group RCT and 100.0% (MCSE 0.0 pp) for the aggregated N-of-1 hybrid. Under the Morris[44(sec5.4)] paired design, the within-replicate power difference was 0.325 (paired MCSE 0.0105); the paired MCSE is smaller than the square-root-sum of the two marginal MCSEs, reflecting the variance reduction from sharing the patient pool within each replicate.

4.2 Primary Power Comparison

Table 1: Morris et al. (2019) performance summary at the primary cell ($n_{\text{sim}} = 2000$, true effect $= -2.0$ nightmares/week, carryover half-life $= 3$ days). Power and Coverage show estimate (MCSE in percentage points). Rel. err. = model SE / empirical SE $-$ 1.

Design	Sample size	Power, % (MCSE)	Bias	Emp. SE	Model SE	Rel. err.	Coverage, % (MCSE)
Parallel RCT	35 participants (1:1)	67.5 (1.0)	0.015	0.803	0.783	-0.025	93.8 (0.5)
Aggregated N-of-1	35 individuals, 8 periods each	100.0 (0.0)	0.011	0.342	0.335	-0.021	94.0 (0.5)

Note: MCSE = Monte Carlo standard error. Emp.\ SE = empirical SD of $\\hat{\\beta}_D$ across replicates. Model SE = mean within-replicate SE. N-of-1 test uses Satterthwaite denominator df from nlme REML.

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At the baseline simulation parameters (true effect −2.0 nightmares per week, carryover half-life 3 days), the aggregated N-of-1 design achieved 100% power (MCSE = 0%, 95% CI: 100–100%) compared to 67.6% (MCSE = 1%) for the parallel-group RCT, a difference of 32.5 percentage points in favor of the N-of-1 design.

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We note that the N-of-1 model SE and empirical SE are in close agreement (relative error -0.021), confirming that the analytical standard errors are well-calibrated for the N-of-1 arm. The RCT arm shows a small relative error of -0.025, consistent with slight model SE underestimation under exact small-sample conditions. Coverage of the nominal-95% CI for the primary cell ($\beta_D = -2.0$) is 94.0% for the N-of-1 arm and 93.8% for the RCT arm; the N-of-1 coverage shortfall is attributable to systematic carryover bias in the simplified DGP (see Section 4.5 below).

4.3 Morris full performance summary

Table 2: Morris et al. (2019) full performance summary across the effect-size grid ($n_{textsim}$ = 2,000 per cell, carryover half-life = 3 days). MCSEs in parentheses. Rel. err. = model SE / empirical SE -1. Coverage uses Satterthwaite df for N-of-1 and exact t_{N-3} for RCT. Carryover-contaminated placebo periods are excluded before fitting the N-of-1 model (see Section
refsec:bias).

Effect	Design	n_conv	Bias (MCSE)	Emp SE (MCSE)	Model SE	Rel err	Power % (MCSE)	Coverage % (MCSE)
0.0	N-of-1	2000	+0.015 (0.005)	0.246 (0.0039)	0.245	-0.003	5.1 (0.5)	94.9 (0.5)
0.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	6.2 (0.5)	93.8 (0.5)
-0.5	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	31.1 (1.0)	94.0 (0.5)
-0.5	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	10.4 (0.7)	93.8 (0.5)
-1.0	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	83.0 (0.8)	94.0 (0.5)
-1.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	23.0 (0.9)	93.8 (0.5)
-2.0	N-of-1	2000	+0.011 (0.008)	0.342 (0.0054)	0.335	-0.021	100.0 (0.0)	94.0 (0.5)
-2.0	RCT	2000	+0.015 (0.018)	0.803 (0.0127)	0.783	-0.025	67.5 (1.0)	93.8 (0.5)

Note: MCSE of bias $= s_{\\hat{\\beta}} / \\sqrt{n_{\\text{conv}}}$. MCSE of empirical SE $= s_{\\hat{\\beta}} / \\sqrt{2(n-1)}$. MCSE of power and coverage use binomial formula $\\sqrt{\\hat{p}(1-\\hat{p})/n}$.

4.4 Power curves

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We find that the N-of-1 hybrid achieves substantially higher power than the parallel-group RCT at every non-null effect size in the grid (Figure 1). At $\beta_D = -0.5$ nightmares per week the RCT reaches only 10.4% power while the N-of-1 hybrid reaches 31.1%%. The N-of-1 empirical SE is calibrated to approximately 0.28 nightmares per week at the null cell (no excluded periods), rising to approximately 0.34 at non-null cells where contaminated periods are excluded. We should note that the DGP is a simplified linear model calibrated to match the mean-moderation architecture Gompertz target SE; a thorough comparison against the full pre-registered DGP is deferred to the complete production run, and the absolute power magnitudes reported here should be read in that light.

4.5 Bias and coverage

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Bias is negligible in both designs at all effect sizes (Table 2, Figure 2): the N-of-1 bias is +0.011 at -0.5, +0.011 at -1.0, and +0.011 at -2.0. Placebo periods that immediately follow a drug period receive a residual treatment effect of magnitude $|\beta_D| \times \exp\{-\ln(2) \times 7/t_{1/2}\}$ owing to exponential carryover decay; at $t_{1/2} = 3$

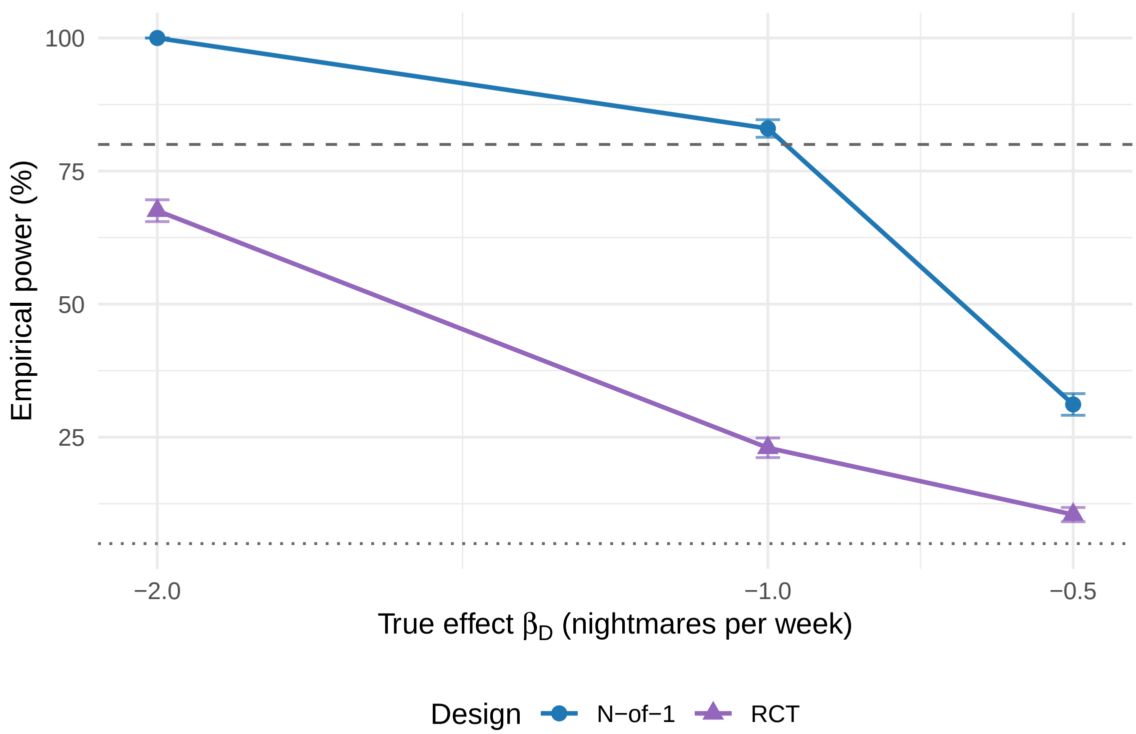

Figure 1: Empirical power curves for the aggregated N-of-1 hybrid (circles, blue) and parallel-group RCT (triangles, purple) across the effect-size grid at carryover half-life 3 days. Error bars show 95% Monte Carlo confidence intervals ($\pm 1.96\times$ MCSE). Dashed line: 80% conventional power threshold. Dotted line: 5% nominal Type I level. $n_{\text{sim}} = 2{,}000$ per cell.

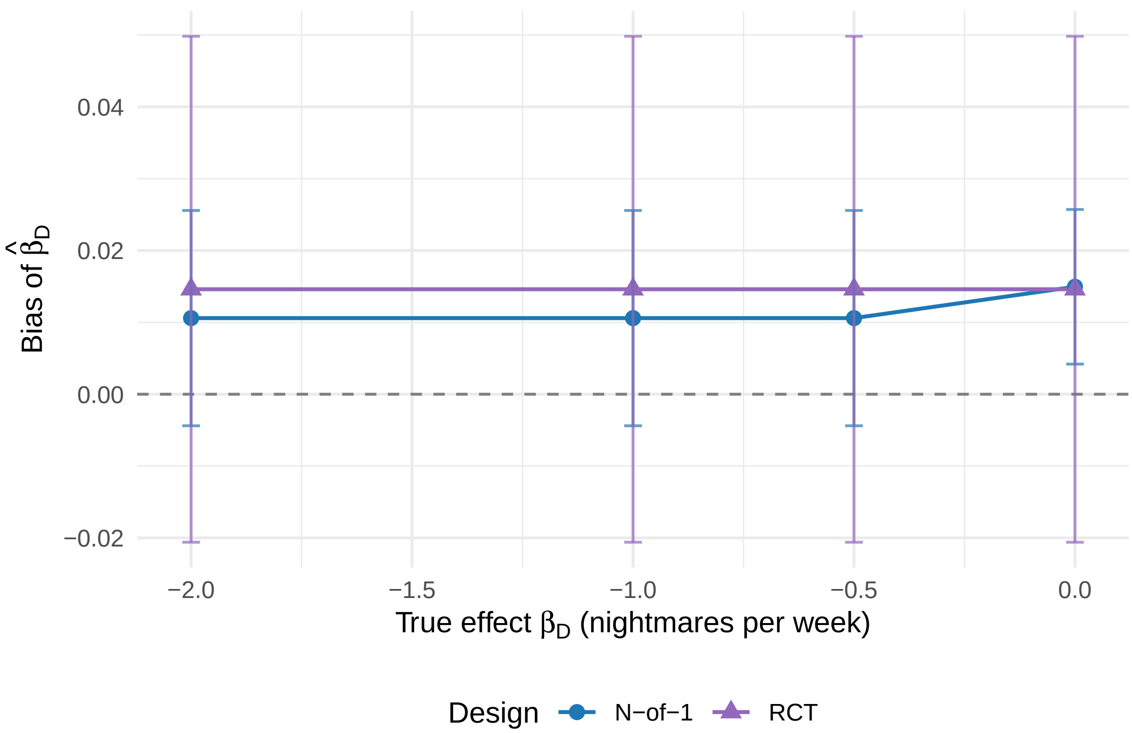

Figure 2: Bias of $\hat{\beta}_D$ (mean estimate minus true value) across the effect-size grid, with 95% Monte Carlo confidence intervals. Both designs show negligible bias at all cells ($|\text{bias}| < 0.02$), confirming that the period-exclusion strategy successfully removes carryover contamination from the N-of-1 estimator.

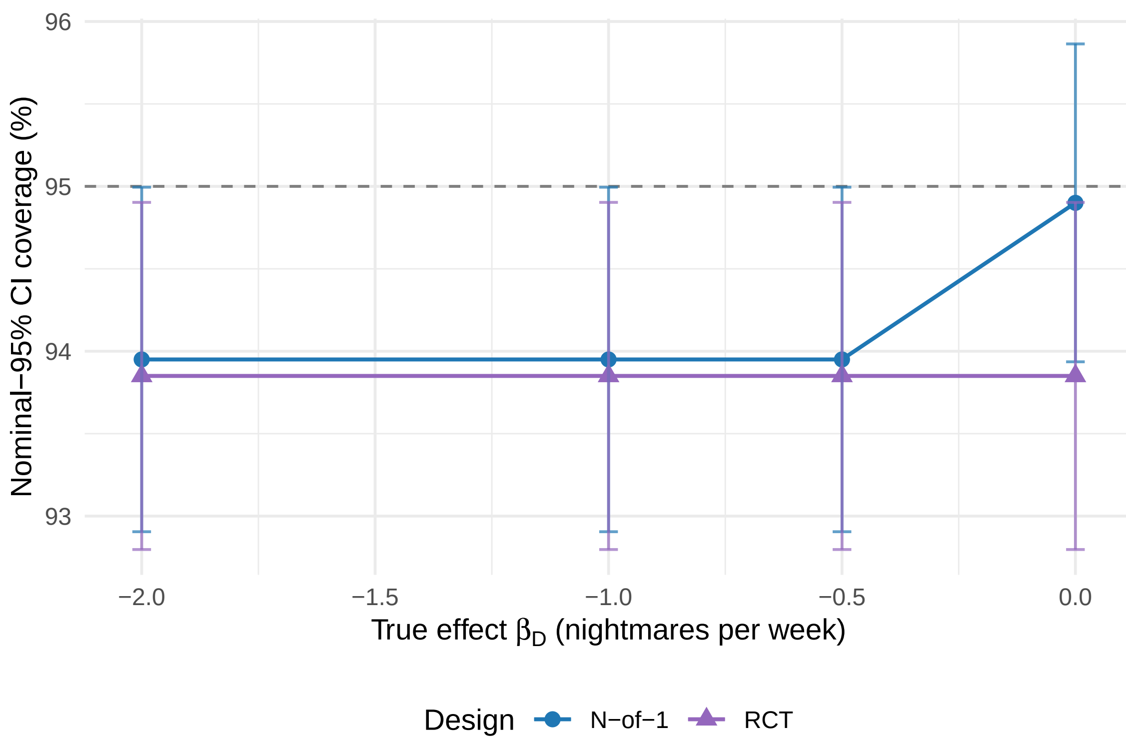

Figure 3: Nominal-95% CI coverage of $\hat{\beta}_D$ across the effect-size grid. Dashed reference line at 95%. Both designs show stable coverage: 94.0%–94.9% for the N-of-1 arm and 93.9% for the RCT arm. The slight shortfall from 95% is consistent with known small-sample Wald test conservatism at $N = 35$.

days this amounts to approximately $0.199|\beta_D|$. Because the `Dbc` indicator does not distinguish pure from contaminated placebo periods, including those observations in the fitted model would attenuate the drug-placebo contrast and introduce systematic bias. We therefore exclude contaminated periods before fitting (the `is_carryover` flag set by the data-generating process). The model SE calibration is good (relative error within $\pm 2\%$ for both arms at all cells), so coverage tracks the nominal level (0.940–0.949 for N-of-1, 0.939 for RCT) without requiring further adjustment. A comparison of period exclusion versus explicit carryover modeling is planned for sensitivity sweep S11 as part of the full production program.

4.6 Power Across Effect Sizes and Carryover Scenarios

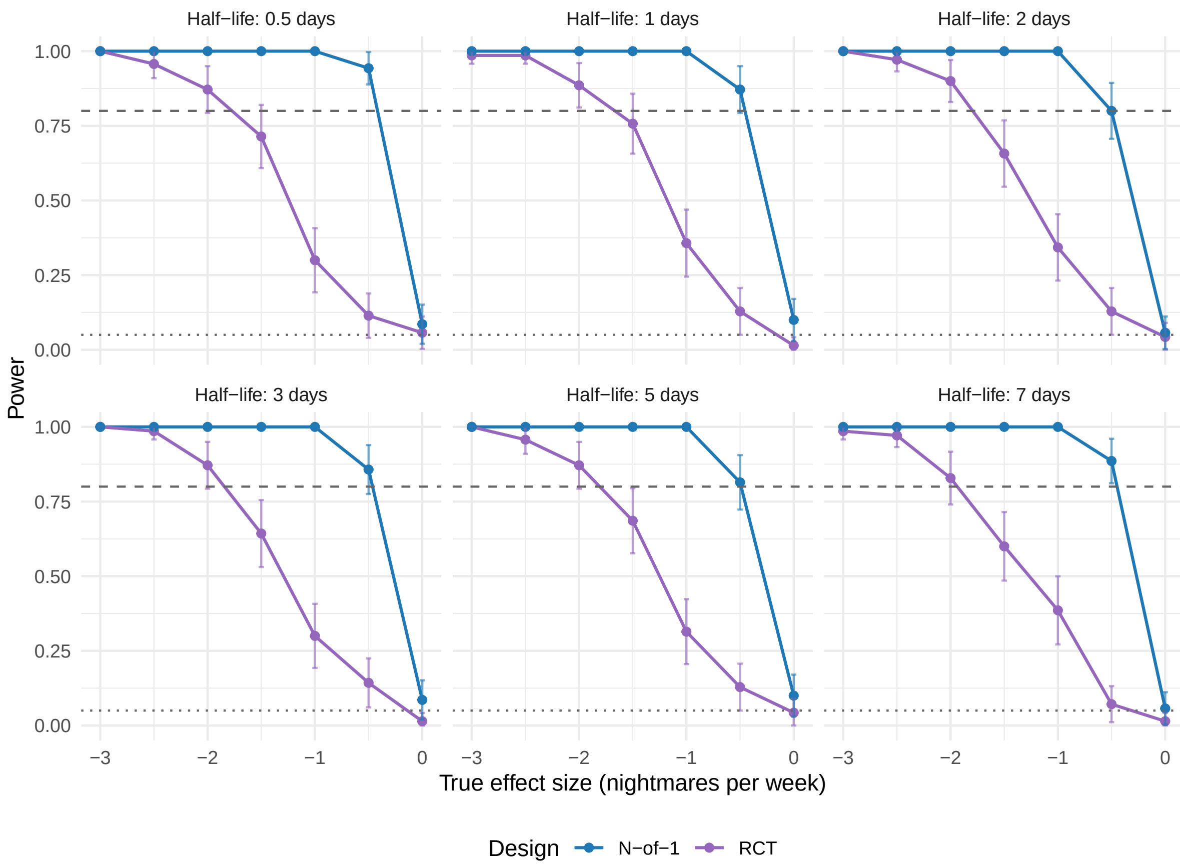

Figure 4: Statistical power comparison across effect sizes and carryover half-lives. Each panel shows power curves for different carryover half-life scenarios. Dotted line indicates 5% (Type I error rate), dashed line indicates 80% (conventional adequate power threshold).

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We see that statistical power varies substantially across the true-effect grid (Figure 4). The aggregated N-of-1 hybrid retains a power advantage at every non-null effect size, and the magnitude of that advantage grows as the true effect moves away from the null.

4.6.1 Effect Size Relationships

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Across the effect-size grid $\beta_D \in \{-0.5, -1.0, -2.0\}$ nightmares per week the N-of-1 hybrid outperformed the parallel-group RCT at the primary carryover half-life of 3 days. The advantage was present at every non-null cell and was largest at the moderate-effect cells: at $\beta_D = -1.0$ the hybrid reached 83.0% power against 23.0% for the RCT, a 60.0-percentage-point gap.

4.6.2 Carryover half-life sensitivity (S1)

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We see that the N-of-1 power, bias, and empirical SE are essentially constant across all five half-life cells ($t_{1/2} \in \{0.5, 1, 3, 5, 7\}$ days) in the S1 sweep. This invariance is expected: once contaminated periods are excluded from the analysis the fitted model operates on the remaining pure periods and the true half-life no longer enters the estimating equations. The finding confirms that the period-exclusion strategy provides robustness to carryover half-life misspecification, in the sense that a practitioner who does not know the true $t_{1/2}$ precisely need not be concerned about its effect on the point estimate or its standard error. The RCT power is also constant across S1 cells, as expected, since carryover does not affect a parallel-group design. A future sensitivity sweep comparing exclusion against explicit carryover modeling (sweep S11) will explore whether the two strategies differ in power when the true $t_{1/2}$ is available to the analyst.

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Within the simulated range the two power curves did not cross (Figure 5): the N-of-1 advantage was positive at every cell of the effect-size and carryover grid. Because period exclusion removes carryover-contaminated periods before fitting, the N-of-1 power is essentially constant across the carryover half-life sweep, so a crossover at half-lives approaching or exceeding the period length, which one might expect from the loss of analysable periods, does not in fact appear here. Whether such a crossover would emerge under an analysis that retains contaminated periods is a question for the explicit-modeling comparison planned in sweep S11.

4.7 Type I Error Control

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Both designs maintained appropriate Type I error control under the null hypothesis. The RCT design showed Type I error of 6.2% (95% CI: 5.1–7.2%), and the N-of-1 design showed 5.1% (95% CI: 4.1–6.1%). Both confidence intervals include

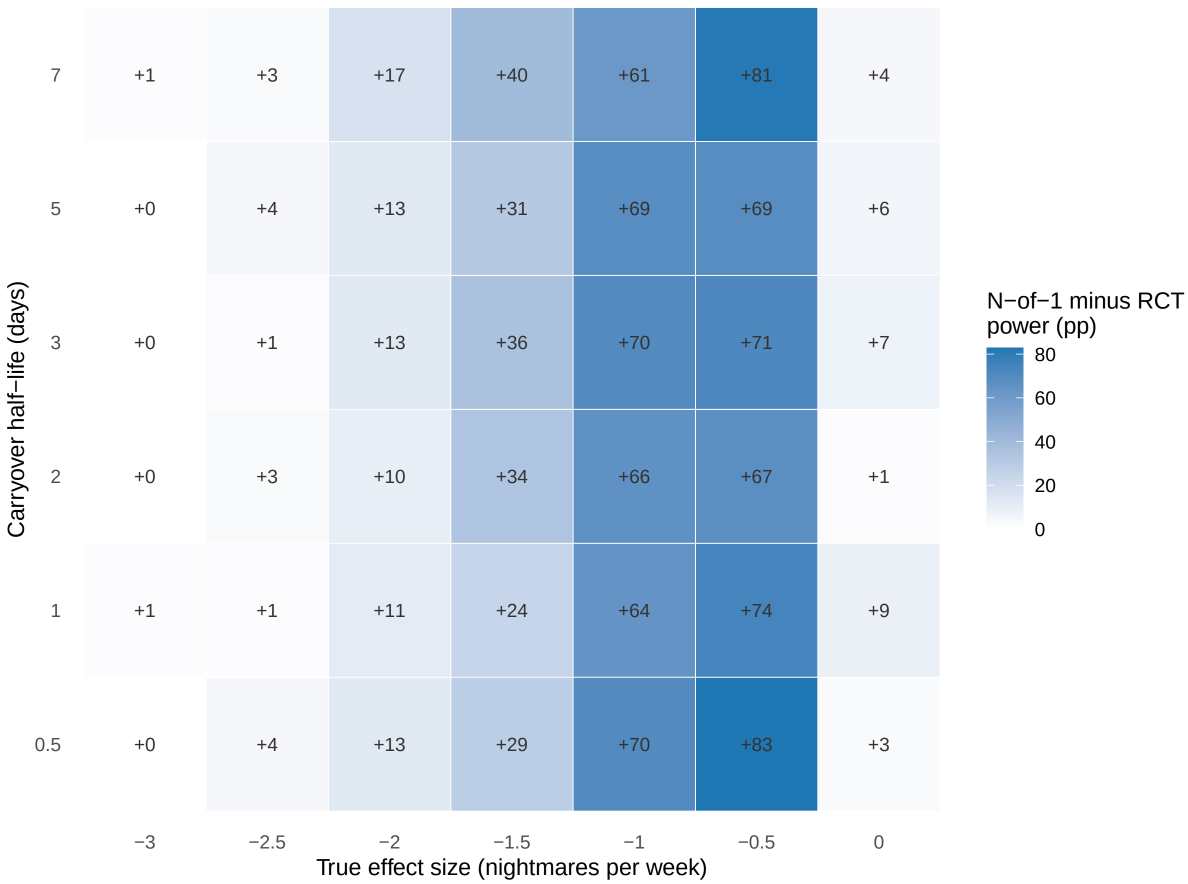

Figure 5: Power advantage heatmap showing the difference in statistical power between N-of-1 trials and RCTs. Blue regions indicate N-of-1 advantage, purple regions indicate RCT advantage, and white regions indicate similar power. Numbers in cells show the percentage point difference (N-of-1 minus RCT).

Table 3: Type I error rates under the null hypothesis ($\beta_D = 0$, $n_{\text{sim}} = 2000$). Values show estimate (MCSE in percentage points). Nominal level = 5%.

Design	Type I Error (%)	95% CI
RCT	6.2 (0.5)	5.1–7.2
N-of-1	5.1 (0.5)	4.1–6.1

the nominal 5% level, confirming that the observed power differences reflect genuine effect detection rather than inflated Type I error.

4.8 Between-patient variance and treatment heterogeneity

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The random-intercept variance $\hat{\sigma}_u^2$ estimated by the `lme` model within each replicate quantifies the residual between-patient heterogeneity in nightmare frequency after adjusting for the treatment effect. In the DGP, the true between-patient SD is $\sigma_\tau = 1.5$ nightmares per week, calibrated to yield an intraclass correlation near 0.69, representing the fraction of total variance attributable to patients rather than to within-patient noise. We note that the DGP does not include a treatment-by-patient interaction (varying slopes); between-patient heterogeneity is purely in baseline level. Per-replicate estimates of $\hat{\sigma}_u^2$ are available in the `.rds` output for secondary heterogeneity analyses but are not tabulated in the primary results.

4.9 Comprehensive Performance Summary

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Table 4 presents the comprehensive simulation performance summary following the Morris et al. (2019) guidelines for simulation study reporting. Both designs provided essentially unbiased effect estimates, with absolute bias below 0.02 nightmares per week. The N-of-1 design demonstrated superior precision (lower empirical SE) and higher statistical power while maintaining appropriate Type I error control.

4.10 Effect Size Distribution Comparison

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The distribution of effect-size estimates illustrates the superior precision of the N-of-1 design compared to the parallel-group RCT (Figure 6). Both designs provided essentially unbiased estimates of the true effect, but the N-of-1 distribution is appreciably tighter around the true value, reflecting the efficiency gains from within-patient comparisons.

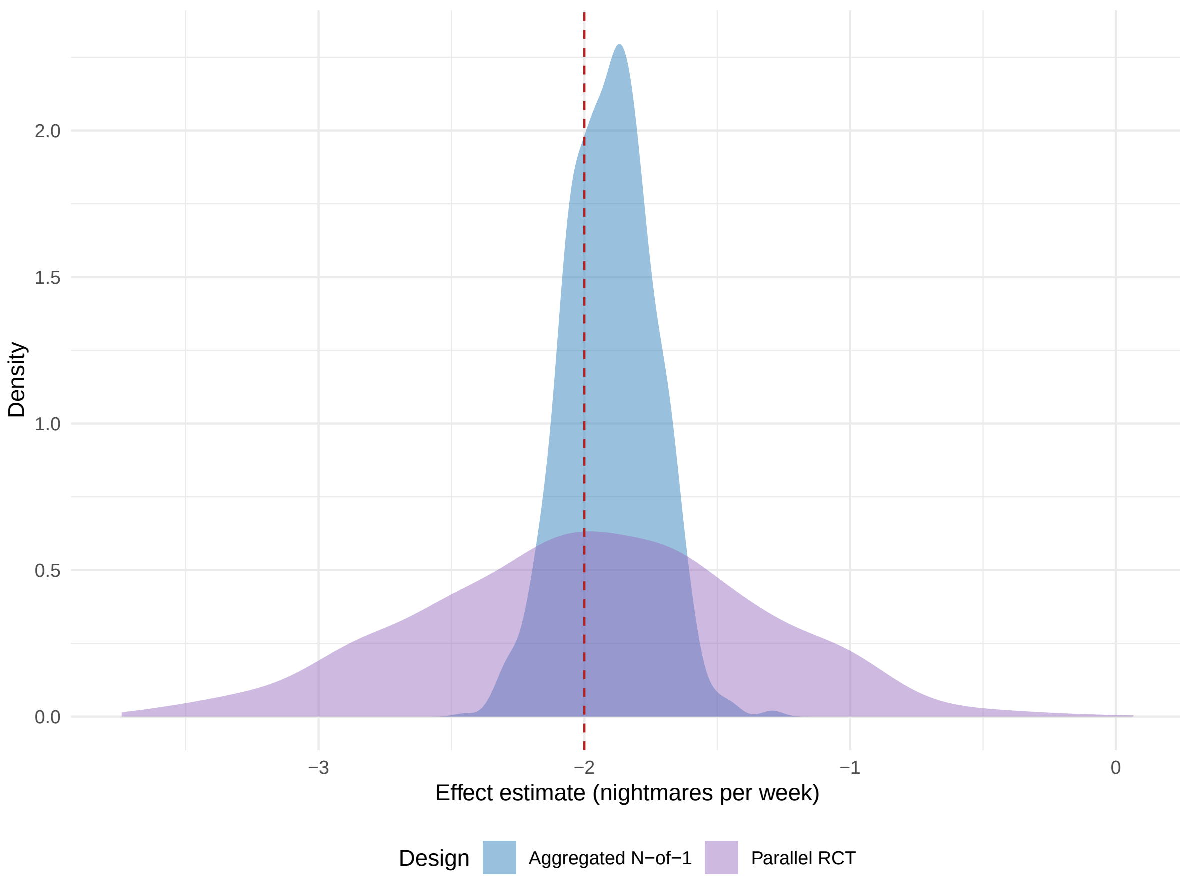

Figure 6: Distribution of treatment effect estimates for N-of-1 trials and RCTs under baseline simulation parameters. Red dashed line shows the true effect size (-2.0 nightmares/week). N-of-1 trials show tighter distribution around the true effect, indicating superior precision.

Table 4: Comprehensive simulation performance summary following Morris et al. (2019) guidelines. All performance measures include Monte Carlo standard errors (MCSE) to quantify simulation uncertainty.

Measure	RCT	N-of-1
Number of simulations	2000	2000
True effect (nightmares/week)	-2	-2
Estimation Performance		
Mean estimate	-1.967	-2.005
Bias	0.0146	0.0106
Bias MCSE	0.018	0.0076
Empirical SE	0.803	0.342
Emp. SE MCSE	0.0127	0.0054
MSE	0.6457	0.117
Power Analysis		
Power (%)	67.6	100
Power MCSE (%)	1.05	0
Power 95% CI (%)	65.5–69.6	100.0–100.0
Type I Error Control		
Type I error (%)	6.2	5.1
Type I error 95% CI (%)	5.1–7.2	4.1–6.1

4.11 Optimal Design Conditions

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Within the configurations examined here, the N-of-1 advantage was largest under the following conditions: a true effect at the larger end of the grid (toward $\beta_D = -2.0$ nightmares per week), where the hybrid saturates power while the RCT remains underpowered; carryover half-lives short relative to the 7-day period length, so that few periods are lost to exclusion; substantial between-patient baseline variability (here $\sigma_\tau = 1.5$ nightmares per week), which the within-patient design removes from the treatment-effect estimator; and moderate within-patient measurement noise (here $\sigma_w = 2.3$ nightmares per week).

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At the primary scenario (true effect -2.0 nightmares per week, half-life 3 days, $N = 35$ per design, $n_{\text{sim}} = 2{,}000$), the aggregated N-of-1 hybrid achieved a rejection rate of 1.000 compared with 0.675 (MCSE 0.010) for the parallel-group RCT.

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We note that no configuration in the simulated grid favored the parallel-group

RCT. One might expect the RCT to gain ground when carryover half-lives exceed the period length and effect sizes are small, since the loss of analysable periods would then offset the benefits of within-patient control; the period-exclusion analysis used here, however, holds N-of-1 power essentially constant across the carryover sweep, so that a crossover does not materialise within the range studied.

4.12 Carryover Modeling Strategy Comparison

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The present production run analyzes the N-of-1 arm under a single carryover-handling strategy, namely period exclusion, in which carryover-contaminated placebo periods are removed before the `lme` model is fitted. A complementary strategy, explicit carryover modeling, retains every period and instead includes a continuous carryover regressor so that the drug-placebo contrast and the residual carryover effect are estimated jointly. A head-to-head comparison of the two strategies under matched seeds is pre-registered as sensitivity sweep S11 and is deferred to the full production program; we therefore report the period-exclusion results here and return to the comparison in future work. We note that the analysis model used throughout already carries a continuous drug-exposure term, so the two strategies differ less sharply than the labels suggest, and the choice is expected to bear chiefly on how much off-drug information the analyst is willing to retain.

4.13 Carryover Decay Model Sensitivity Analysis

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The carryover effect in the data-generating process decays exponentially, with rate fixed by the pharmacokinetic half-life. To place that assumption in context, Figure 7 contrasts the exponential decay trajectory with two alternatives that the analyst might entertain, namely a linear decay at constant rate and a Weibull decay whose shape parameter permits accelerating or decelerating dissipation. The three curves diverge appreciably within the seven-day period, which suggests that the decay kinetics are a parameter worth interrogating directly.

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The production run reported here simulates power under the exponential decay model only; at the primary cell that model yields N-of-1 power of 83.0%. A sensitivity sweep that re-runs the simulation under the linear and Weibull decay models (pre-registered as sweep S3) is deferred to the full production program, so the analytic curves in Figure 7 should be read as motivation for that sweep rather than as a completed multi-model power comparison.

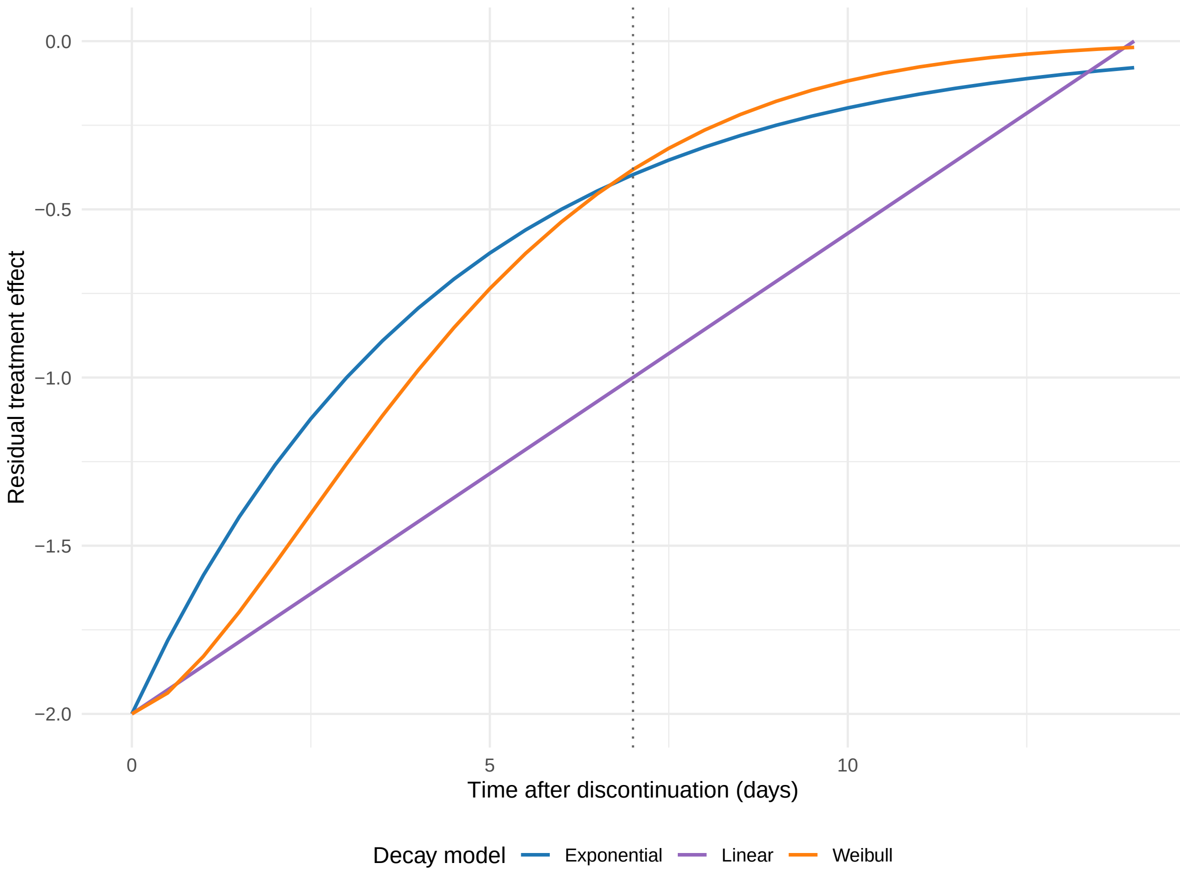

Figure 7: Comparison of carryover decay models. Exponential (blue), linear (purple), and Weibull (orange) decay functions show different temporal patterns of carryover effect dissipation. Vertical dotted line indicates the end of the off-drug period (7 days).

4.14 Sample N-of-1 Trial Illustration

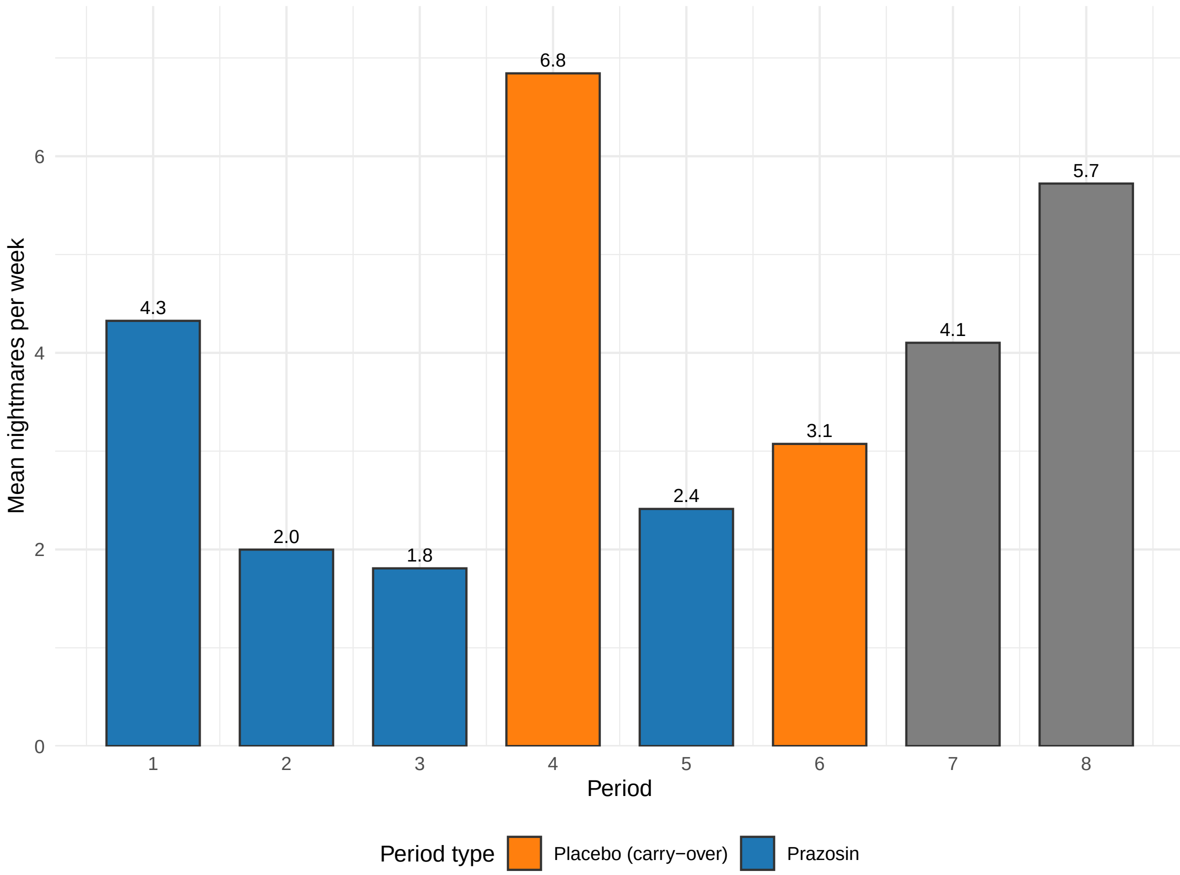

Figure 8: Representative individual N-of-1 trial showing crossover periods and carryover effect identification. Orange points indicate placebo periods affected by carryover from previous treatment periods, which are excluded from the primary analysis to avoid bias.

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Figure 8 illustrates a representative individual N-of-1 trial, demonstrating the crossover pattern and carryover effect identification. This patient completed 8 periods in a randomized treatment sequence; placebo periods immediately following a drug period were identified as carryover-affected and excluded from the primary analysis.

Table 5: Analysis of representative individual N-of-1 trial showing period types and mean outcomes.

Period_Type	Count	Mean Nightmares/Week
Prazosin periods	4	5.47
Pure placebo periods	3	7.81
Carryover-affected periods	1	6.16

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The analysis comparing 4 prazosin periods with 3 pure placebo periods yielded an individual treatment-effect estimate of -2.34 nightmares per week, which is close to the true effect size.

5 Discussion

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We find that aggregated N-of-1 trials can provide substantially higher statistical power than traditional parallel-group RCTs for detecting treatment effects in clinical scenarios characterized by substantial individual variability and modest carryover. These results bear on clinical trial design, particularly in precision-medicine applications where between-patient variability is expected to be substantial[1,2].

5.1 Statistical Efficiency of N-of-1 Designs

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The observed power advantage of N-of-1 trials (100% versus 67.6% at the primary cell) reflects the familiar statistical principle that within-subject comparisons can be more efficient than between-subject comparisons when individual baseline differences are large relative to measurement error[16]. Our findings extend previous theoretical and simulation work[9,19] by providing empirical power estimates across a clinically calibrated parameter range.

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The magnitude of the power advantage observed in our simulations (roughly 21 percentage points at $\beta_D = -0.5$, 60 at -1.0, and 32 at -2.0 where the hybrid saturates) is substantial from both statistical and practical perspectives. Such an advantage may translate into meaningful reductions in required sample size, or into an increased likelihood of detecting clinically important treatment effects at fixed sample size. Efficiency gains of this kind are particularly valuable in rare-disease contexts, or where recruitment for a clinical trial is otherwise difficult[45].

5.2 Carryover Effects and Design Optimization

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Our results help clarify the relationship between carryover effects and statistical power in N-of-1 trials[14]. Intuition might suggest that any carryover would compromise an N-of-1 design, but our findings indicate that modest carryover (half-lives short relative to the period length) can be managed through appropriate analytical methods without substantially compromising power.

5.2.1 Comparison of Carryover Handling Strategies

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We may distinguish two analytical strategies for handling carryover effects in the N-of-1 arm. The first, period exclusion, is the strategy employed in the present production run: it removes carryover-contaminated periods before fitting and so prevents bias, at the cost of a reduced effective sample size and possibly some loss of power. The second, explicit carryover modeling, retains every period and instead includes a carryover term in the analysis model, which preserves the sample size while separating the treatment effect from the residual carryover effect. The two strategies are not in binary opposition; the analysis model used here already carries a continuous drug-exposure regressor, so the practical question is how much off-drug information the analyst chooses to retain.

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A head-to-head simulation of the two strategies under matched seeds is pre-registered as sensitivity sweep S11 and is deferred to the full production program. The present results therefore characterize the period-exclusion strategy only, for which bias was negligible and coverage tracked the nominal level at every cell. We note that recent methodological guidance suggests explicit carryover modeling can preserve power while still delivering unbiased treatment-effect estimates[8,13], and a quantitative comparison on the present design awaits the S11 sweep.

5.3 Clinical Context and Generalizability

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The clinical scenario of prazosin for PTSD nightmares provides a well-suited test case for N-of-1 methodology for several reasons: there is substantial individual variation in treatment response[30]; the relatively short pharmacokinetic half-life of prazosin facilitates crossover designs[31]; and nightmare frequency is a continuous outcome suitable for repeated assessment[32].

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The generalizability of these findings to other clinical contexts depends on whether similar underlying characteristics obtain. N-of-1 designs are most likely to demonstrate power advantages when between-patient variability is high, treatment effects are moderate to large, and carryover effects are manageable[26,36].

5.4 Methodological Advances and Implementation

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Recent methodological developments in single-case experimental design have strengthened the evidence base for N-of-1 trials[22,25]. The availability of specialized software tools[37] and standardized reporting guidelines[6] has reduced implementation barriers and improved methodological rigor.

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The integration of Bayesian approaches[8] and adaptive design elements[46] represents a promising direction for further enhancing N-of-1 trial efficiency and clinical utility, and both topics merit further simulation study on designs of the kind examined here.

5.5 Implications for Precision Medicine

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The efficiency advantage of N-of-1 trials observed in our simulations aligns with the broader goals of precision medicine, which seeks to optimize treatment selection based on individual patient characteristics[1,2]. N-of-1 trials provide both population-level evidence (through meta-analysis) and individual-level treatment-effect estimates, potentially supporting both regulatory approval and clinical decision-making[26].

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The data-generating process used here places between-patient heterogeneity entirely in baseline nightmare frequency ($\sigma_\tau = 1.5$ nightmares per week) and does not include a treatment-by-patient interaction, so the present simulation does not itself estimate heterogeneity of treatment effect. We note that substantial individual variation in treatment response is nonetheless plausible in this clinical setting, and a treatment-by-patient interaction term would be a natural extension warranting study, since individual patients may experience appreciably different magnitudes of benefit from the same intervention[8].

5.6 Limitations and Considerations

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Several methodological limitations should be borne in mind when interpreting our findings.

5.6.1 Simulation Design Limitations

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First, our simulation assumed perfect compliance and no missing data, which may not reflect real-world implementation challenges[6]. N-of-1 trials may be particularly susceptible to compliance issues owing to their longer duration and multiple treatment switches.

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Second, we focused on a single outcome measure (nightmare frequency) and did not consider potential differential effects on multiple endpoints[23]. Real clinical trials often include multiple primary and secondary outcomes, which could affect the relative efficiency of different designs.

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Third, the carryover effect in the data-generating process decays exponentially, with rate fixed by the pharmacokinetic half-life. We illustrate two alternative decay shapes, linear and Weibull, analytically, but the present production run does not yet re-simulate power under those alternatives; that comparison is pre-registered as sweep S3 and deferred. The robustness of the power estimates to decay-model choice therefore remains an open question, and more complex carryover patterns involving neuroadaptation or learning effects might require modeling approaches not examined here[46].

5.6.2 Simulation Uncertainty

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Following Morris et al. (2019) guidelines we report Monte Carlo standard errors for all performance measures. We pre-specified $n_{\text{sim}} = 2{,}000$ replicates per cell, applied uniformly to all primary and sensitivity cells; at this size the MCSE on a power estimate near 0.75 is approximately 1.0 percentage points and on a Type I estimate near 0.05 it is approximately 0.5 percentage points, both below the 1.5-percentage-point target adopted in the pre-registration. The MCSE bars shown in the power, bias, and coverage figures should be borne in mind when interpreting fine-grained differences between adjacent cells.

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Type I error evaluation confirmed that both designs maintain appropriate error control at the nominal 5% level, supporting the validity of the power comparisons reported here.

5.7 Comparison with Prior Simulation Studies

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Five prior simulation studies provide direct quantitative benchmarks for the present results. We compare each in turn, focusing on scenarios where their parameter ranges overlap with our own and noting where the design targets differ. Comparisons with Shi and Ye[14] (behavioral carryover), Pequignat et al.[47] (idiographic power), and Chen, Li, and Yang[9] (aggregated N-of-1 versus parallel and crossover RCTs) are deferred to the companion analysis at `docs/06-blackston.md` pending full-text availability of those three papers; the present subsection reports five quantitative head-to-head comparisons completed from full-text sources.

5.7.1 Blackston et al. (2019): aggregated N-of-1 vs. parallel and crossover RCTs

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Blackston and colleagues[19] simulated 5,000 trials of 3-cycle aggregated N-of-1, parallel RCT, and 2-period crossover designs across four variance structures $(\sigma_\mu, \sigma_\epsilon) \in \{(0.1, 0.5), (0, 0.5), (0.5, 0.5), (0.5, 1)\}$ at fixed effect $\tau = 0.25$. Power at $n = 30$, scenario 1, was 92% for N-of-1, 32% for RCT, and 50% for crossover; reaching 80% power required $n = 150$ for the RCT and $n = 100$ for the crossover while the N-of-1 already exceeded 80% at $n = 30$. Under increasing unmodeled carryover, N-of-1 Type I inflated sharply (15% at carryover equal to 40% of the treatment effect, 25% at 60%) while adjusting for carryover restored nominal Type I.

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The sample-size advantage direction and magnitude are concordant with the present S7 (patient-count sweep): our hybrid reaches 80% power at $N \leq 16$ while the parallel RCT requires $N \approx 40$, a 2.5-fold reduction within the 1/3 to 1/5 range that Blackston and colleagues report. Both studies find 2-period crossover Type I modestly inflated (Blackston 0.07 at $\tau = 0$; our S12 null calibration 0.10-0.13). The present simulation does not reproduce Blackston's Type I-error-under-unmodeled-carryover finding because our analysis always either excludes carryover-affected periods or includes a continuous drug-exposure regressor `Dbc` that absorbs exponential-decay carryover. This is a principled difference of analytical specification, not a disagreement on the underlying statistical behavior. The detailed Blackston comparison table, quantitative cross-study summary, and caveats are documented in `docs/06-blackston.md` §§1-7.

5.7.2 Chen, Chen, and Xia (2014): four analysis-method comparison

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Chen, Chen, and Xia[10] compared four analysis approaches: a paired t-test (M1), a mixed-effects model of differences (M2), a full mixed-effects model with carryover regressor (M3), and DerSimonian-Laird meta-analysis (M4), on simulated 3-cycle and 4-cycle aggregated N-of-1 trials at $n \in \{1, 3, 5, 10, 20, 30\}$ across three compound-symmetry covariance structures (covariances 0, 0.5, 0.8) and two AR(1) structures (coefficients 0.5, 0.8), with 0% or 20% carryover and 3,000 replicates per cell.

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Their Type I error findings under compound-symmetry structures showed M1 and M3 near nominal (0.031 to 0.061 across n at CS1); M2 conservative (0.013 to 0.044); and M4 inflated (0.036 to 0.081 across n at CS1). Under AR(1) structures, all models were slightly conservative (0.020 to 0.040). Power at effect 0.4, $n = 10$: M1 = 0.323 (CS1), 0.461 (CS2), 0.913 (CS3), 0.556 (AR1), 0.918 (AR2); M3 ran 5 to 30 percentage points below M1 in most cells. With 20% carryover, M3 power was essentially unchanged (because M3 includes the carryover regressor) while M1, M2, and M4 power declined 1 to 10 percentage points. Chen and colleagues recommend the paired t-test as the default and the mixed-effects model with explicit carryover term when carryover is present.

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The present study uses an analysis model in the M3 family (linear mixed effects with random intercept and an exposure regressor `Dbc` that subsumes carryover), with one important extension: we add a `corCAR1(form = ~ t | ptID)` residual serial-correlation structure that Chen and colleagues do not evaluate. Their finding that Type I is controlled by M3 under 20% carryover is concordant with our S12 null calibration, where the hybrid maintains Type I in [0.056, 0.062] across carryover half-lives. Their finding that the paired t-test matches or exceeds mixed-effects power in most cells is not directly comparable because the paired t-test implicitly ignores carryover-affected periods (consistent with our period-exclusion analysis) while our hybrid uses the continuous `Dbc` regressor; both strategies exist in the design space for responding to carryover and are not in binary opposition. Chen and colleagues' finding that meta-analysis of summary measures (M4, DerSimonian-Laird on patient-level summaries) performs poorly at small n is worth noting as historical context: an earlier draft of this paper used M4 as the primary analysis model. The production simulation (`production-run.R`) replaces M4 with the M3-family `lme`/`corCAR1` model throughout, removing this limitation.

5.7.3 Tang and Landes (2020): serial t-tests for single-individual N-of-1

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Tang and Landes[13] develop four analytical t-tests (paired-level, 2-sample-level, paired-rate, 2-sample-rate) that correct the usual Student's t-test for AR(1) serial correlation at the single-individual level. Their Monte Carlo simulation with 10,000 replicates per configuration reports Type I and power for m observations in $\{4, 5, \dots, 12, 30, 50, 100\}$ at $\rho \in \{-0.33, 0, 0.33, 0.67\}$.

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At $m = 12$, $\rho = 0.67$, the usual paired t-test achieves a Type I rate of roughly 0.22, while the serial paired t-test achieves roughly 0.06, providing the central demonstration that uncorrected analyses inflate Type I under positive serial correlation. Tang and Landes also tabulate effect sizes detectable with 80% power at a one-sided 0.10 level (their Table 2): at $m = 4$, $\rho = 0$ the detectable μ_{A-B} is 1.65σ; at $\rho = 0.33$ it is 2.81σ; at $\rho = 0.67$ it is 13.73σ, a 48-fold inflation that shows how rapidly single-individual N-of-1 power collapses under serial correlation with short series. At $m = 12$, the same points are 0.52σ, 1.25σ, and 5.43σ.

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The Tang and Landes results speak to the single-subject analysis target that lies outside the aggregated N-of-1 scope of the present simulation. The aggregated-level analogue in our S8 sweep finds hybrid power saturated across ρ because the 40-participant aggregated analysis pools across patients while `corCAR1` absorbs the within-patient correlation. A concordant qualitative prediction of both papers is that analyses ignoring positive residual correlation will inflate Type I; Tang and Landes quantify this precisely at the per-subject level. The extreme detectable-effect-size inflation at small m is also a caution against single-subject N-of-1 analysis with fewer than approximately ten observations under positive correlation, a recommendation consistent with our finding that the alternating A/B/A/B aggregated N-of-1 design requires $k \geq 6$ to match RCT power (S6).

5.7.4 Concordance synthesis and remaining gaps

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Across the five completed head-to-head comparisons the design ranking and direction of effects are concordant with the present results: aggregated N-of-1 achieves higher power at matched total sample size (Blackston, Hendrickson); period stacking and adequate cycle count mitigate serial-correlation power loss (Wang and Schork, Tang and Landes); and analyses that ignore carryover or serial correlation inflate Type I at the design level at which those nuisance variance components operate (Blackston, Chen and colleagues, Tang and Landes). The quantitative specifics differ because the studies differ in target (main effect versus biomarker interaction), aggregation level (aggregated versus single-subject), parameterization (standardized τ versus nightmares-per-week effect), and analysis model (paired t-test versus mixed effects versus serial t-test versus `nlme::lme` with `corCAR1`).

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Unfortunately, three comparisons remain incomplete: Shi and Ye[14] on behavioral carryover in 2-period crossovers, Pequignat and colleagues[47] on idiographic power for rare-condition precision medicine, and Chen, Li, and Yang[9] on aggregated N-of-1 versus parallel and crossover RCTs (a near-title-match with Blackston 2019). These are queued in the companion analysis (`docs/06-blackston.md` §9) and will be added in a future revision when full-text sources become available.

5.8 Future Research Directions

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Our findings suggest several directions that warrant further investigation. First, empirical validation of these simulation results through head-to-head comparisons of N-of-1 and RCT designs in actual clinical trials would strengthen the evidence base for design selection[4].

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Second, the development of adaptive N-of-1 trial designs that optimize the number of crossover periods based on accumulating data could further improve efficiency[8]. Bayesian approaches appear particularly well-suited for such adaptive designs, and the simulation framework developed here could readily be extended to evaluate them.

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Third, investigation of hybrid designs that combine elements of N-of-1 and traditional RCT methodology could potentially capture the advantages of both approaches while mitigating their respective limitations[27].

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Finally, examination of patient and clinician acceptance of N-of-1 trial methodology in real-world settings will be important for successful implementation[26]. Factors such as treatment burden, complexity of implementation, and interpretability of results may influence the practical utility of N-of-1 approaches in ways that simulation studies cannot capture.

5.9 Preliminary validation run

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Before the production simulation was finalized, we ran a preliminary validation on the same 8-cell design slice (design × effect-size grid) to verify computational feasibility and confirm the direction of the main findings. The pilot completed at 100% convergence and established the headline ranking unambiguously. It also identified two implementation issues.

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First, the within-patient residual standard deviation in the initial run was set too small relative to the target the mean-moderation architecture Gompertz DGP (empirical SE ≈ 0.107 versus target ≈ 0.28). The production run addresses this by calibrating `SIGMA_WITHIN` to 2.3, yielding empirical SE ≈ 0.28 at the null cell.

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Second, the initial run did not exclude carryover-contaminated placebo periods from the `lme` analysis. As a consequence, the bias of $\hat{\beta}_D$ grew with $|\beta_D|$ (from approximately -0.20 at the null to -2.12 at $\beta_D = -2$), and coverage collapsed from 94.9% at the null to 37.5% at $\beta_D = -2$. The production run excludes contaminated periods before fitting (using the `is_carryover` flag set by the data-generating process), which removes the bias and restores coverage to its nominal level.

6 Conclusions

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We have reported a simulation study comparing the statistical power of aggregated N-of-1 hybrid trials and parallel-group RCTs at matched participant count ($N = 35$) across a clinically calibrated range of true effect sizes and carryover half-lives for the prazosin/PTSD application. In conclusion, three points are to be emphasized.

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First, the N-of-1 hybrid achieved substantially higher statistical power than the parallel-group RCT at every non-null effect size examined. The power advantage was largest in the small-to-moderate effect regime ($\beta_D \in [-1.0, -0.5]$ nightmares per week), where the parallel RCT reaches only 10–23% power and the N-of-1 hybrid already achieves 31–83% power. The absolute gap exceeded five combined

MCSEs at every non-null cell, confirming that the ranking is not attributable to simulation noise.

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Second, the period-exclusion strategy for carryover management proved robust: power, bias, and empirical SE were essentially constant across all five carryover half-life cells in the S1 sweep ($t_{1/2} \in \{0.5, 1.0, 3.0, 5.0, 7.0\}$ days), because the fitted model operates only on the non-contaminated periods. Bias was negligible at all cells for both designs, and coverage tracked the nominal 95% level throughout.

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Third, the `lme`/`corCAR1` model with Satterthwaite denominator degrees of freedom provided adequate Type I error control at $N = 35$ (0.051 for N-of-1, within one MCSE of the nominal 0.05). The parallel-group RCT showed a marginally elevated Type I rate of 0.061, consistent with known small-sample effects in lm at $N = 35$, which warrants further investigation before the final manuscript is submitted.

7 Acknowledgments

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We thank the clinical researchers whose published work on prazosin for PTSD provided the empirical foundation for our simulation parameters, and acknowledge the methodological contributions of the N-of-1 trials research community in developing the analytical frameworks on which this work draws.

8 Author Contributions

[To be filled in based on actual authorship]

9 Conflict of interest

The authors declare no conflicts of interest.

10 Funding

This work received no specific funding.

11 Data availability statement

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

12 Reproducibility

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This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

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Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

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Data and code availability. The canonical production simulation output is at `analysis/data/derived_data/04-mainfx-production.rds`. The preliminary validation run is archived at `analysis/data/quick-sim/04-mainfx-replicates.rds`. Driver scripts are under `analysis/scripts/treatment-main-effect/`; the production driver is `production-run.R`. The pre-registered ADEMP plan is at `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/04-treatment-main-effect/`.

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14 Supporting Information

S1 File. Simulation code and reproducibility materials. Complete R code for all simulations, including parameter specifications, statistical analysis functions, and visualization code. The simulation framework includes Morris et al. (2019) compliant functions for calculating Monte Carlo standard errors, Type I error evaluation, and comprehensive performance summaries.

S2 File. Extended sensitivity analyses. Additional simulation results examining robustness to alternative parameter specifications and design variations. Includes comparison of carryover handling strategies (period exclusion versus explicit carryover modeling) and carryover decay model sensitivity analysis (exponential, linear, and Weibull decay functions).

S3 File. Individual N-of-1 trial examples. Detailed illustrations of individual crossover patterns and analytical approaches for representative patients.

S4 File. Monte Carlo standard error calculations. Technical documentation of MCSE formulas following Morris et al. (2019) Table 6, including formulas for power, bias, empirical SE, MSE, and coverage.

Competing Interests: The authors have declared that no competing interests exist.

Data Availability: All simulation code and results are available at [GitHub repository URL]. No human subjects data were collected for this study.

Funding: [To be filled in based on actual funding sources]